A security bound from the original QEF operators

Mathematical derivation, source assumptions, and corrected tables

The numerical benchmark is conditional; the original source law is not certified.

1 Security model and a rotation-moment bound

The security quantity considered here is the number of independently encrypted quantum preparations needed to distinguish two specified images. This is a sample-complexity statement, rather than a gate-complexity estimate for exhaustive key search. The proof uses the original three QEF layers. It does not assume that an adversary reconstructs an arbitrary unitary or recovers every component of the original key.

Keys, ciphertexts, and the attack experiment. Write A for the substitution-angle material produced by the nD-QHCS and $B = (K^{\mathrm{perm}}, K^{\mathrm{ent}})$ for the material produced by the HGΦ-QRNG. These are two source classes. The two components of B are described by their joint distribution; independence is not inferred from their output lengths. Each encryption uses a fresh secret key record, and the receiver obtains the same record over an assumed secure channel. Independent refresh between preparations is an explicit security assumption, stronger than merely obtaining different numerical key values. The experiment gives the adversary no additional side information correlated with the target secret record; any claimed extension to source leakage must bound that leakage explicitly.

Let $N = HW$, let r denote the feature-register size, and let $D = N2^r$ denote the dimension of the valid image subspace. Unused padded position addresses are excluded from the ciphertext experiment. The image states have the form

$$|I\rangle = N^{-1/2} \sum_{k=0}^{N-1} |k\rangle|\chi_k\rangle, \qquad \langle\chi_k|\chi_k\rangle = 1. \tag{1}$$

The adversary is given the classical descriptions of two such images of the same public dimensions. A uniform hidden bit b selects I_b. The adversary receives q encryptions of that selected image, with independently refreshed keys, and may perform any joint quantum measurement. The publicly transmitted ciphertext is the quantum state, not a classical list of all its amplitudes or a rendered image file. Define

$$\sigma_b = \mathbb{E}_{A,B}[U_{A,B}|I_b\rangle\langle I_b|U_{A,B}^\dagger]. \tag{2}$$

The joint ciphertext is $\sigma_b^{\otimes q}$. Its optimal binary discrimination probability is [1]

$$P_{\mathrm{guess}}^{(q)} = \frac{1}{2} + \frac{1}{4}\left\|\sigma_0^{\otimes q} - \sigma_1^{\otimes q}\right\|_1. \tag{3}$$

This experiment does not give repeated access to one fixed-key unitary. It also does not cover key reuse, arbitrary reference-entangled chosen inputs, authentication, or publication of simulator amplitude records.

Use the actual circuit structure. For the intended pixel-dependent substitution, write

$$R_A = \sum_k |k\rangle\langle k| \otimes R_{A,k}, \qquad R_{A,k} = \bigotimes_{j=1}^{r} T_{k,j}, \quad T_{k,j} = R_z(\theta_{k,j}^z)R_y(\theta_{k,j}^y)R_x(\theta_{k,j}^x). \tag{4}$$

Position-controlled CNOT/CZ gates leave each position basis label unchanged. Thus, for suitable feature unitaries $V_B(t)$,

$$E_B = \sum_t |t\rangle\langle t| \otimes V_B(t), \qquad U_{A,B} = E_B P_B R_A = \sum_k |\pi_B(k)\rangle\langle k| \otimes V_B(\pi_B(k))R_{A,k}. \tag{5}$$

The permutation and entanglement keys may have any joint distribution in the theorem below. In particular, a uniform distribution over all $N!$ permutations is not needed for the image experiment in Eq. (1). The permutation must nevertheless be implemented reversibly with its work registers correctly cleaned.

An explicit angle reference law. The following law is an analytical reference, not an experimentally certified property of the nD-QHCS. For an integer $L \geq 2$, let each Euler angle be uniform on

$$\Theta_L = \{2\pi s/L : s = 0, \ldots, L-1\}. \tag{6}$$

Conditioned on B, the $3r$ angles within a pixel are independent, and the angle tuples belonging to any two different pixels are independent. Pairwise independence of pixel tuples is sufficient; independence of all N tuples jointly is not required. The continuous reference is the corresponding independent uniform law on $[0, 2\pi)$. These conditions must hold after conditioning on any public setup information or side information used in the attack experiment. The reference is a key-law assumption, not a consequence of deterministic seed expansion. For $N \geq 2$, two independent uniform angle tuples alone require at least $6r \log_2 L$ bits of conditional source entropy. The continuous law is an ideal limiting model; a finite implementation must use the finite-law expression and provide sufficient actual key material.

Set

$$a_L = \frac{1}{L \sin(\pi/(2L))}, \qquad \beta_L = a_L^{6r}, \qquad \beta_\infty = \left(\frac{2}{\pi}\right)^{6r}. \tag{7}$$

Uniform Euler coordinates are not Haar-distributed rotations. The nonzero first moment is retained in the analysis below.

Theorem 1 (Conditional image-discrimination sample bound)**.** *Under the stated angle reference law and independent key refresh, let $N \geq 2$, $c_N = 1 - 1/N$, and*

$$C_{N,r,L} = \ln\left(1 + D\beta_L^2 c_N\right). \tag{8}$$

For every image state in Eq. (1)*, its average ciphertext σ satisfies*

$$\frac{1}{2}\|\sigma - I_D/D\|_1 \leq \beta_L c_N, \tag{9}$$

$$\mathsf{D}(\sigma \| I_D/D) \leq C_{N,r,L}, \tag{10}$$

where D uses natural logarithms. For any two such images, achieving $P_{\text{guess}}^{(q)} \geq p > 1/2$ requires

$$\boxed{q \;\geq\; \left\lceil \frac{\ln 2 - h(p)}{\ln(1 + N2^r \beta_L^2 (1 - 1/N))} \right\rceil, \qquad h(p) = -p \ln p - (1-p)\ln(1-p).} \tag{11}$$

The continuous reference uses β_∞ in the same expressions. For $N = 1$, the feature twirl makes the reference ciphertext exactly maximally mixed, and no finite number of fresh encryptions distinguishes the candidate images.

Proof. For one rotation axis a, averaging its conjugation over Θ_L dephases the Bloch components transverse to that axis. Composing the independent x, y, and z averages therefore gives

$$\mathbb{E}[TXT^\dagger] = \frac{I_2}{2} \operatorname{Tr} X. \tag{12}$$

For the first moment, a finite trigonometric sum gives

$$\mathbb{E}[R_a(\theta)] = c_L I_2 - i s_L \sigma_a, \quad c_L = 1/L, \quad s_L = L^{-1} \cot(\pi/(2L)). \tag{13}$$

Each matrix on the right is a_L times a unitary, since $c_L^2 + s_L^2 = a_L^2$. Hence $M = \mathbb{E}[R_{A,k}] = \beta_L^{1/2} W$ for a feature unitary W, and the feature conjugation average is the completely depolarizing channel. In the continuous limit,

$$\mathbb{E}[T] = (2/\pi)^3 I_2, \qquad M = (2/\pi)^{3r} I_{2^r}. \tag{14}$$

For example, the first identity follows by multiplying $\mathbb{E}[R_a] = -i(2/\pi)\sigma_a$ in the order z, y, x.

Write $\rho = |I\rangle\langle I| = \sum_{k,l} |k\rangle\langle l| \otimes X_{kl}$, and let Δ remove its off-diagonal position blocks. Conditional pairwise independence gives

$$\mathbb{E}[R_{A,k} X_{kl} R_{A,l}^\dagger] = \begin{cases} I_{2^r} \operatorname{Tr}(X_{kk})/2^r, & k = l, \\ M X_{kl} M^\dagger, & k \neq l. \end{cases} \tag{15}$$

For Eq. (1), $\operatorname{Tr} X_{kk} = 1/N$. Consequently the diagonal contribution is $\tau = I_D/D$, which is invariant under every $E_B P_B$. The remaining term is

$$\sigma - \tau = \beta_L \, \mathbb{E}_B[F_B(\rho - \Delta\rho)F_B^\dagger], \quad F_B = E_B P_B (I \otimes W). \tag{16}$$

Every F_B is unitary, so both trace norm and Hilbert–Schmidt norm are bounded by the corresponding norms before this average.

The pinching map is an average over N diagonal position-phase unitaries, one of which is the identity. The triangle inequality gives $\|\rho - \Delta\rho\|_1 \leq 2(1 - 1/N)$. Combining this with Eq. (16) proves Eq. (9). Moreover, because ρ is pure and its position probabilities are uniform,

$$\|\rho - \Delta\rho\|_2^2 = \operatorname{Tr}\rho^2 - \operatorname{Tr}(\Delta\rho)^2 = 1 - 1/N. \tag{17}$$

Thus $\|\sigma - \tau\|_2^2 \leq \beta_L^2 c_N$. Jensen's inequality applied to the eigenvalues of σ yields

$$\begin{aligned} \mathsf{D}(\sigma\|\tau) &= \ln D + \operatorname{Tr}(\sigma \ln \sigma) && (18) \\ &\leq \ln(D \operatorname{Tr}\sigma^2) = \ln(1 + D\|\sigma - \tau\|_2^2) \leq C_{N,r,L}. && (19) \end{aligned}$$

For a uniform hidden bit and q independent encryptions, the Holevo information obeys

$$\chi_q \leq \frac{1}{2} \sum_{b=0}^{1} \mathsf{D}(\sigma_b^{\otimes q} \| \tau^{\otimes q}) \leq q C_{N,r,L}. \tag{20}$$

Indeed, subtracting χ_q from the first expression gives the nonnegative relative entropy of the average ciphertext to $\tau^{\otimes q}$. Holevo's theorem and binary Fano's inequality [1] imply that any guess with success at least p requires mutual information at least $\ln 2 - h(p)$. Therefore $\ln 2 - h(p) \leq q C_{N,r,L}$, proving the result. The same argument allows any joint measurement and unlimited computational postprocessing. □

Numerical meaning and source certification. For the QIRBP feature count $r = 11$, the continuous reference gives $\beta_\infty \simeq 1.13785812768644 \times 10^{-13}$. Table 1 evaluates Eq. (11) at $p = 2/3$. The numbers are necessary ciphertext-preparation counts under the specified law; they are neither measured attack times nor bit-security exponents. The logarithm of a sample count must not be labelled as elementary-gate security.

Table 1: Conditional QIRBP sample lower bounds for binary image discrimination. Independent uniform Euler angles on $[0, 2\pi)$, fresh independent keys, $r = 11$, and success $p = 2/3$ are assumed. The actual nD-QHCS/HGΦ source conditions remain uncertified.

Image	N	$D = N2^{11}$	Necessary preparations q
Ashurbanipal	65,536	134,217,728	$\geq 3.25904 \times 10^{16}$
Hammurabi	50,394	103,206,912	$\geq 4.23831 \times 10^{16}$
Lamassu	65,536	134,217,728	$\geq 3.25904 \times 10^{16}$
Babylon Lion	786,432	1,610,612,736	$\geq 2.71583 \times 10^{15}$

To relate the reference calculation to an actual generator, suppose the resulting average ciphertext is within trace distance κ of the reference ciphertext, uniformly over the allowed input images. For example, a bound κ on the total variation distance of the actual joint finite angle/control-key law from its finite reference law

suffices. Comparing a finite law directly to a continuous law in total variation is inappropriate; the reference must use the same finite alphabet. Then the preceding proof gives

$$C_\kappa = \ln\left(1 + D[\beta_L\sqrt{1-1/N} + 2\kappa]^2\right), \qquad q \geq \left\lceil \frac{\ln 2 - h(p)}{C_\kappa} \right\rceil. \tag{21}$$

The trace-distance bound becomes $\frac{1}{2}\|\sigma - \tau\|_1 \leq \beta_L(1 - 1/N) + \kappa$. This follows from $\|\sigma - \sigma_{\text{ref}}\|_2 \leq \|\sigma - \sigma_{\text{ref}}\|_1 \leq 2\kappa$. The source bound must include correlations between pixels, axes, source parameters, both components of B, and any relevant side information. Statistical image metrics or balanced one-bit marginals do not certify this joint bound.

The mapping $\theta = \gamma \bmod (q, \pi)$ does not by itself establish the reference law. Even if $q \bmod \pi$ were uniform, its angle range would be $[0, \gamma\pi)$; the reference continuous range requires $\gamma = 2$, as well as the stated joint moment conditions. No measured value of κ or proof of those conditions for the original generators is supplied here. Therefore the numerical rows are conditional benchmarks and are not asserted as certified resistance of the experimental generator.

Why the range cannot be omitted. There is an explicit contrasting calculation within the same circuit family. Suppose the Euler angles are instead independent uniform variables on $[0, \pi)$, independently of B. Averaging the Bloch rotations gives, for a data qubit initially in $|0\rangle$ or $|1\rangle$,

$$\langle X\rangle_0 = (2/\pi)^2, \qquad \langle X\rangle_1 = -(2/\pi)^2. \tag{22}$$

Consider all-black and all-white QIRBP images. At output position $t = 0$, no position control is active, so $V_B(0) = I$. That position occurs with probability $1/N$, regardless of the permutation. An adversary can request fresh encryptions until measuring $t = 0$, then measure X on one colour-data qubit. This procedure uses N preparations on average and succeeds with probability

$$\frac{1 + (2/\pi)^2}{2} \simeq 0.702642 > 2/3. \tag{23}$$

It is a conditional countermodel, not a claimed attack on an unspecified nD-QHCS law. It shows mathematically why one cannot assign a source-independent numerical resistance to the three-layer architecture or to the statement that keys are refreshed.

Interpretation of the original attack labels. Independent known/chosen-image pairs from other operations do not share the target key under the independent-refresh assumption. They can still reveal a biased generator law or shared source parameters; the conditional key law must therefore be specified. Generic key search, if a reversible candidate verifier is available, has the usual conditional $\Theta(\sqrt{M/t})$ query baseline for M candidates and t marked candidates [2, 3, 4]. No particular old key-length value is used to certify QEF, and no reduction proving that all attacks require exhaustive search is claimed. The dimension or parameter count of a general quantum state or unitary is not substituted for an attack lower bound.

Theorem 1 concerns the mathematical quantum operators and the stated key law. Classical array simulations and their rendered images do not verify these quantum moment conditions. In particular, the original scalar-seeded XOR/permutation simulation is not an implementation of the controlled complex Euler rotations assumed here. Image-quality and decryption metrics are therefore reported as diagnostics, rather than as evidence that the conditional security bound holds. The new probabilistic bound is proved analytically; it is not included in the existing Lean 4 unitarity checks. Entanglement alone supplies neither authentication nor a proof of resistance to active modification.

2 Verification and finite-angle benchmarks

The finite reference law uses L equally spaced angles in $[0, 2\pi)$; L is not the number of image intensity levels. The table lists the necessary independently encrypted preparations at success $2/3$ for $r = 11$. Integer thresholds were calculated at 100 and 140 decimal digits and agree. These are theorem evaluations, not executed cryptanalytic attacks.

Table 2: Additional conditional finite-angle sample lower bounds.

Image	$L = 2$	$L = 4$	$L = 256$
IM1	$\geq 3.1134 \times 10^{10}$	$\geq 1.0765 \times 10^{15}$	$\geq 3.2563 \times 10^{16}$
IM2	$\geq 4.0490 \times 10^{10}$	$\geq 1.3999 \times 10^{15}$	$\geq 4.2348 \times 10^{16}$
IM3	$\geq 3.1134 \times 10^{10}$	$\geq 1.0765 \times 10^{15}$	$\geq 3.2563 \times 10^{16}$
IM4	$\geq 2.5945 \times 10^{9}$	$\geq 8.9708 \times 10^{13}$	$\geq 2.7135 \times 10^{15}$

Table 3: Verification of the mathematical derivation on small systems.

Check	Result
Exact angle reference models	4
Enumerated-channel image checks	80
Joint binary measurement checks	10
Source-perturbation cases	12
Normalized QIRBP image checks	6
Maximum first-moment error	1.17×10^{-16}
Maximum feature-twirl error	1.11×10^{-16}
Maximum enumerated-channel error	1.8×10^{-16}
Original source certification	Not established

The small-system channel calculation enumerates all substitution keys on four models and compares the resulting superoperator with the derived moment formula. It checks the full channel, rather than only the norm of a few outputs. The image checks then verify trace norm, Hilbert–Schmidt norm, and relative-entropy inequalities. Joint binary measurements are evaluated directly on small product states and checked against the Holevo/Fano converse. Separate perturbation cases check the source-error inequality. These checks support the algebra but cannot certify an unspecified HCS source distribution.

3 Impact on the manuscript and remaining work

The revision replaces the old quantum-attack subsection and its "passed" table. Its abstract, discussion, comparisons, and conclusion are adjusted to the proved scope. Pixel-dependent angle indexing and the chronological Euler/inverse order are made consistent with the operator used by the theorem. The receiver uses the delivered stochastic key values rather than drawing new QRNG outcomes. Hardware statements distinguish small IBM circuits from the large-image simulations.

The three encryption layers and both proposed original key-source classes remain in the mathematical architecture. No QOTP layer or v2 uniform privacy core is used in this derivation. A quantum-safe confidentiality claim for the actual original source still requires a certified finite joint angle/control law, the actual range parameter, an imperfection bound, and implementation correspondence. The sorting work-register cleanup must also be checked independently; the theorem assumes an ideal clean address permutation. The new proof has not been formalized in Lean 4.

A particularly important consequence is the range countermodel in Section 1. With the independent uniform $[0, \pi)$ reference, the expected sample counts for the demonstrated postselection attack are only $65,536$, $50,394$, $65,536$, and $786,432$ for the four images, at success 0.702642. Those are attack upper bounds in

that contrasting law; they are not the theorem's lower bounds and are not measured attacks on the original generator. The sharp difference prevents assigning one of the reference numbers to the actual system before its angle law is established.

References

[1] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. Chapters 3 and 5, including Theorem 5.49 and Lemma 5.54. `https://cs.uwaterloo.ca/~watrous/TQI/`.

[2] L. K. Grover, A fast quantum mechanical algorithm for database search, STOC 1996.

[3] M. Boyer, G. Brassard, P. Høyer, and A. Tapp, Tight bounds on quantum searching, Fortschritte der Physik 46 (1998). `https://arxiv.org/abs/quant-ph/9605034`.

[4] C. Zalka, Grover's quantum searching algorithm is optimal, Physical Review A 60, 2746–2751 (1999). `https://arxiv.org/abs/quant-ph/9711070`.